

LyzrNegotiate: Autonomous B2B Negotiation Platform

Official System Documentation & Architecture Guide

1. Executive Overview

LyzrNegotiate is a production-grade, multi-agent AI platform designed to automate B2B vendor and procurement negotiations. By leveraging Lyzr Agent Studio, the platform pits an autonomous AI Buyer against an autonomous AI Vendor to negotiate contract terms (Price, Delivery Days, and SLA Penalties) in real-time. This eliminates the weeks of email ping-pong typically required for enterprise B2B contracting.

2. Primary Use Cases

This platform is designed for enterprise procurement and supply-chain teams. Key use cases include:

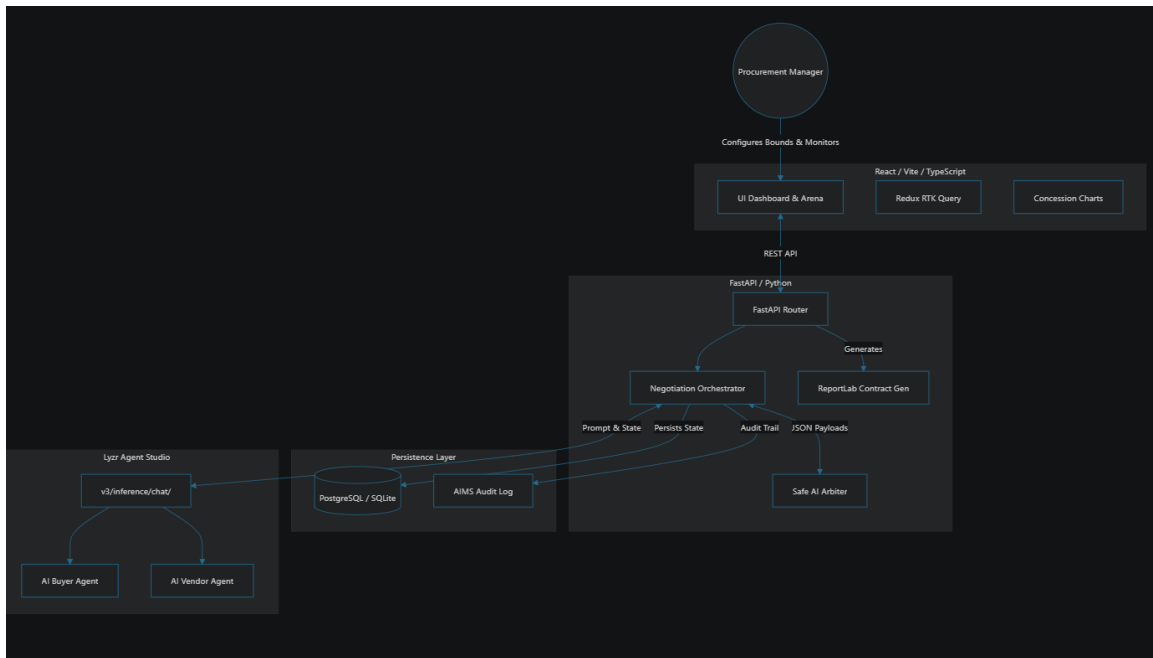
- **Procurement Automation:** Automatically haggle with software vendors, freelancers, or raw material suppliers to secure the best possible price without human intervention.
- **Dynamic Vendor Quoting:** Allow clients to immediately negotiate with your company's AI sales agent, providing instant concessions and finalizing contracts 24/7.
- **Safe AI Contracting:** Ensure that no AI agent ever agrees to a deal outside of strict financial limits, thanks to the mathematical guardrails enforced by the Safe AI Arbiter.
- **Tamper-Proof Audit Trails:** Automatically log every AI decision, concession, and justification to a persistent database and the AIMS governance layer for compliance tracking.

3. System Architecture

The application is decoupled into three primary layers ensuring security, scalability, and ease of deployment. The architecture follows a strict separation of concerns:

Data Flow:

1. The User configures policy bounds in the React Frontend.
2. The Frontend sends a REST API request to the FastAPI Backend.
3. The FastAPI Orchestrator initiates a negotiation loop.
4. The Orchestrator calls the Lyzr Agent Studio API to get responses from the AI Buyer and AI Vendor.
5. Before recording any bid, the Safe AI Arbiter mathematically validates the JSON payload.
6. Validated bids are saved to the PostgreSQL Database and broadcast back to the Frontend UI.
7. Upon consensus, the backend dynamically generates a legally formatted PDF contract.



4. Component Breakdown

4.1 The Frontend (React + Vite + TypeScript)

The frontend serves as the control center for human oversight, built with TailwindCSS for an enterprise SaaS feel.

- **Setup Dashboard:** Users define strict policy bounds (Max Budget, Min SLA, Max Delivery).
- **Negotiation Arena:** A real-time monitoring room tracking declining price curves via Recharts.
- **History Dashboard:** A centralized view of all past negotiations to download finalized PDF contracts.

4.2 The Backend (FastAPI + Python + SQLAlchemy)

The backend acts as the secure 'referee' between the user, the database, and the AI agents.

- **The Orchestrator:** Manages the turn-based conversation loop and database sessions.
- **Safe AI Arbiter:** A deterministic, mathematical firewall that rejects any AI hallucination or out-of-bounds offer.
- **Contract Generation:** Compiles the agreed-upon price, SLA, and delivery days into a legally formatted PDF using ReportLab.

4.3 The AI Layer (Lyzr Agent Studio)

Connects directly to the official Lyzr Agent Studio inference endpoints (v3/inference/chat/).

- **Personas:** Agents are configured in the Lyzr Studio UI with specific negotiation tactics.
- **JSON Schemas:** The agents are strictly prompted to output parsable JSON containing price, delivery, SLA, and justifications.

5. Cloud Deployment Strategy

This project is optimized for modern Platform-as-a-Service (PaaS) providers to ensure a seamless hackathon deployment and presentation.

- **Frontend:** Hosted on Vercel utilizing the vercel.json routing rewrite for Single Page Applications.
- **Backend & DB:** Hosted on Render using Render's free PostgreSQL managed database to prevent ephemeral disk wipes and guarantee persistence.